

Power Budget Analysis

Assumptions:

- Battery: 11.1V, 1800mAh (19.98 Wh)
- Efficiency of regulators: ~90%
- Operation split: Within 25 min base (navigation), 3 mins active mechanisms

Component	Voltage (V)	Current (A)	Power (W)
TurtleBot Base (Pi + OpenCR + LiDAR)	11.1	0.87	9.66
1x USB Camera	5	0.2	1 (~1.11W input)
Ultrasonic Sensor	5	0.015	0.075 (~0.083W input)
1x RS360 Motor	12	0.8	7.2
L298N Loss (~20%)	12	0.3	3.6
SG90 Servo	5	0.1	0.5 (~0.55W input)

Power Calculations:

Navigation Phase (25 min):

Total Power ≈ 10.77 W

Energy = $10.77 \times (25/60) \approx 4.49$ Wh

Shooting Phase (3 min):

Total Power ≈ 22.203 W

Energy = $22.203 \times (3/60) \approx 1.11$ Wh

Total Energy Consumption ≈ 5.6 Wh

Conclusion:

Battery capacity (19.98 Wh) $\gg$ required (5.6 Wh).

Even accounting for inefficiencies and peaks, 25 min runtime is comfortably achievable.